

Chemistry.

MCQs:-

1. B
2. B
3. B
4. B
5. C
6. B
7. B
8. B
9. B
10. C
11. B
12. B

Q: 2(i)

A: 2(i)

① a) Molar mass of SO_2 :-

$$32 + 2(16) = 64 \text{ mol/g}$$

② b)

$$\text{mass} = \frac{\text{molar mass}}{64} \times \frac{1}{6.022 \times 10^{23}} \times N_A$$

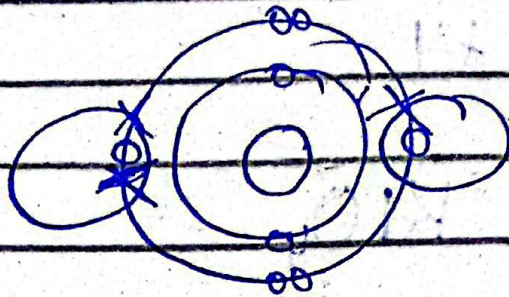
$$\frac{\text{molar mass}}{N_A}$$

$$\frac{64}{6.022 \times 10^{23}} \Rightarrow 1.06 \times 10^{-22} \text{ g}$$

Q: 2(ii)

A: 2(ii)

Diagram :-



types of bonds :-
Covalent bonds

Oxygen has 2 lone pairs of electron.

Q: 2 (iii)
A: 2 (iii)
② S in H_2SO_3
Solution:-

$$2(1) + x + 3(-2) = 0$$

$$2 + x - 6 = 0$$

$$x - 6 + 2 = 0$$

$$x - 4 = 0$$

$$x = +4$$

③ C in Na_2CO_3

$$+1(2) + x + 3(-2) = 0$$

$$2 + x - 6 = 0$$

$$x - 6 + 2 = 0$$

$$x - 4 = 0$$

$$x = +4$$

④ I in KIO_4

$$1 + x + 4(-2) = 0$$

$$1 + x - 8 = 0$$

$$x - 8 + 1 = 0$$

$$x - 7 = 0$$

$$x = +7$$

$$Q: 3(i)$$

$$A: 3(i)$$

① Convert it into 100g of compound.

$$\begin{aligned} \cdot C &= 26.7 \text{ g} & \cdot H &= 2.2 \text{ g} \\ \cdot O &= 71.1 \text{ g} \end{aligned}$$

Convert grams into moles.

$$\text{moles} = \frac{\text{mass}}{\text{molar mass}}$$

$$\begin{aligned} * C &= \frac{26.7}{12} & * H &= \frac{2.2}{1} & * O &= \frac{71.1}{16} \\ &= 2.225 & &= 2.2 & &= 4.443 \end{aligned}$$

Divide into to the smallest value.

$$\textcircled{1} \quad \frac{2.225}{2.2} \approx 1$$

$$\textcircled{2} \quad \frac{2.2}{2.2} = 1$$

$$\textcircled{3} \quad \frac{4.443}{2.2} \approx 2$$

Empirical formula:- SO_2 CHO_2 1:1:2

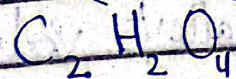
② Empirical formula:- CHO_2
① molar mass of CHO_2

$$12 + 1 + 2(16) \\ 13 + 32 \\ 45 \text{ g/mol}$$

ii) $\text{moles} = \frac{\text{mass}}{\text{molar mass}}$

$$= \frac{90}{45}$$

$$= 2$$



{ molecular formula }

③

moles :- ?

mass :- 18g

molar mass of CHO_2 :- 45g

$$\text{moles} = \frac{\text{mass}}{\text{molar mass}}$$

$$= \frac{18}{45}$$

$$= 0.4$$

$$\frac{18}{45} = 0.4 \text{ mol/g}$$

Q:3 ————— (ii) —————

A:3 ————— (ii) —————

(ii)

(a)

$$\frac{\text{mass n.o of Ag}^{(A)} \times \text{NA of Ag}^{(A)} + \text{mass n.o of Ag}^{(B)} \times \text{NA of Ag}^{(B)}}{100}$$

$$107 \times 52 + 109 \times 48$$

100

$$5564 + 5232$$

100

$$10796$$

100

$$= 107.96$$

(b)

Protons of Ag = 47

Electrons of Ag = 47

neutrons of Ag = $109 - 47$
62

(c) Electronic Configuration of N^7

$$1s^2, 2s^2, 2p^3$$

~~$$1s^2, 2s^2, 2p^3, 1s^2, 2s^2, 2p^3$$~~

group :- 2nd

period :- $2 + 3 + 10 = 15$ or 3